



Katherine van Roon



Dutch-American



vanroonk@gmail.com



www.linkedin.com/in/katherinevanroon



+31 6 3935 4041



<https://k-van-roon.github.io/portfolio/>

Climate Hazards Consultant & Geo-Data Specialist

Expertise in flood risk modelling, anticipatory action, and disaster risk reduction. Experienced in building and maintaining tools and their data dependencies, GIS-based analysis, and delivering operational information products for government and humanitarian clients. Track record leading international projects, coordinating multidisciplinary stakeholders, and translating complex technical outputs into actionable insights for end-users. Latest updates and projects available [here](#).

EXPERIENCE

Climate Hazards Consultant/Developer, Fast Hazard, | March 2025–Present

- Delivered specialized flood modelling, hazard assessments, and exposure analysis to support risk management, planning, and emergency response
- Lead multiple international flood modelling projects for government and humanitarian clients across Canada, Palestine, and Caribbean regions
- Provided emergency flood expertise as a part of Near-Real Time Modelling Team for [Dutch Disaster Risk Reduction and Surge Support \(DRRS\) programme](#), delivering rapid response support including flood mapping and exposure analysis for floods
- Served as Project Lead for “Investigating FastFlood for Regional Modelling Application” (Natural Resources Canada), managing client communications, task delegation, and technical delivery of flood and landslide modelling
- Project Lead for SAFEPLACE consortium (ESA-funded European crisis response initiative), managing team communications and integrating FastFlood API into emergency response hub for rapid flood projections
- Technical manager for [EUROSION](#) project, facilitating design and development of multiscale process-based soil erosion models, focusing on the water erosion model
- Coordinated Mediterranean Prevention, Preparedness, and Response to Disasters ([PPRD-MED](#)) consortium flood projection project for Gaza, Palestine, managing complex stakeholder communications under challenging operational conditions and completing technical flood simulations and exposure projections
- Contributed to software development including Python and Svelte programming for bug fixes, feature additions, and API development
- Managed business development activities including Request for Proposal (RFP) preparation, Contract Change Notice (CCN) submissions for project bids, and open-call project bids
- Supervised development team and assisted with changes to website, branding, and marketing of company
- Acted as technical point of contact for clients, conducting product demonstrations, onboarding session, and troubleshooting support

Intern, Deltares | August–December 2024

- Designed uncertainty estimation module for FloodAdapt software using Python and Monte Carlo simulations
- Created workflow to propagate uncertainty for flood risk estimations and analyzed results using global sensitivity analysis
- Organized and led workshops on uncertainty integration within multidisciplinary teams
- Produced comprehensive technical documentation for uncertainty propagation workflow

SKILLS

Geo-Spatial Skills: Flood (Risk) Modelling (FastFlood, LISEM, HEC-RAS), Exposure Projections & Mapping, Near-Real Time Modelling, GIS Analysis and Mapping (QGIS, ArcGIS), Remote Sensing Analysis (Google Earth Engine), Spatial Data Analysis, Monte Carlo Simulations, Global Sensitivity Analysis

Technical Skills: Python, API Development, Svelte, Cloud Environment (Azure, Cloudflare), Statistical Analysis (R Studio), Data Visualization, Microsoft Power Automate

Soft Skills: Project Leadership, Client Management, Emergency Response Coordination, Business Development, RFP/Proposal Writing, Stakeholder Coordination, Team Management, Research & Analysis, Problem-Solving

Languages: English (Native), Dutch (A2/B1), Spanish (C1)

EDUCATION

2022–2024

MSc Spatial Engineering, University of Twente

- Thesis: [Integrating Probabilistic Weather Forecasting for Flood Early Warning in Dominica](#)
- Specialisation: Natural Hazards and Disaster Risk Reduction

2019–2022

BSc Technology & Liberal Arts and Sciences, University of Twente

- Honour's Program: University College Twente, ATLAS Program